Saransh

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Education

May2025 | **J.C. Bose University of Science and Technology, YMCA, Faridabad** **Faridabad, India**
Dec2021 | B.Tech. in Electronics Engineering with specialisation in Internet of Things

Experience

Jul2025 | **Vivo** **Noida, India**
Mar2025 | *Graduate Engineer Trainee*

- Worked on defect, repair and root cause analysis of new flagship smartphones models.
- Performed board- and component-level diagnostics using oscilloscopes, multimeters, regulated power supplies and software tools to identify faults.
- Received a Pre-placement Full-time offer for my contributions.

Mar2025 | **Ultrasonics Metrology Lab, CSIR-National Physical Laboratory** [] **New Delhi, India**
Jan 2025 | *Undergraduate Researcher | Advisor: Prof. P. K. Dubey*

- Worked on ideas in applied physics and embedded systems, focusing on ToF-based precision sensing.
- Implemented multi-sample averaging and signal processing techniques.
- Our work was accepted at ICUMSAT'25.

Publications

[2025] **Development of a Device for the Estimation of Ambient Temperature by measuring Ultrasonic Propagation Velocity in Air** [Paper |]
Saransh, Nitin Dhiman and P.K. Dubey
International Conference on Ultrasonics and Applied Material Science for Advanced Technology **[ICUMSAT]**

Select Projects

Smart IoT- Based Ventilator System Jul'24 - Nov'24
[EEN-753] | Advisor: Prof. Sunil Jadav

- Designed a real-time embedded monitoring system integrating biomedical sensors with cloud connectivity.
- Achieved functional parity with ~$4,000 commercial systems at a ~$110 build cost.

Railway Line Fault Detection System Aug'24 - Oct'24
[EE-IOT-752] | Advisor: Prof. Rohit Tripathi

- Developed a robot that can autonomously detect obstacles and prevent potential hazards.
- Core implementation included programming the logic to stop the motor and trigger an alert via GSM module when objects are detected within threshold.

Bob, The Obstacle Avoider Jan'24 - Feb'24

- Designed and built an autonomous line-following robot with a multi-channel IR sensor array and PID-based adaptive thresholding for precise trajectory tracking.

Honours and Awards

Samsung, IISc and Synopsys Inc., USA Fellowship, ISWDP, 2026 [] Accepted to attend a 2 months semi-focused program.

CSIR - National Chemical Laboratory, 2025 [] Selected for a fully funded 18 months research project under *Prof. Muthukumarasamy Karthikeyan* in collaboration with IIT Indore in Pune, India.

Robotics Hackathon, BML University, 2024 [] Ranked 1st among 67 teams, winning cash prize worth 10,000 INR.

Technical Skills

Electronics & Design: KiCAD, LTSpice, MATLAB/Simulink **Programming & Verification:** C/C++, Python, VHDL, SystemVerilog, UVM **Microcontrollers & Platforms:** ATmega16A, Arduino, ESP8266, Git **Communication Protocols:** UART, SPI, I2C, CAN, USB, PCIe **Testing & Measurement:** Oscilloscope, Logic Analyzer, Function Generator, Multimeter, LCR Meter, Spectrum Analyzer